Imagine a little girl holding a red balloon in a crowded park. She is with her mother and it is a beautiful spring afternoon. Her mother looks away momentarily, distracted by some noise and when she looks back, the child is gone. She desperately calls out her baby's name and bystanders join in the desperate search. They quickly report the missing child to the information desk in the park, where security officers view surveillance camera feeds—both closest to the location where the girl went missing and across the entire park's camera network, looking for a child holding a red balloon. And within seconds, officers locate the child who had wandered off.

Sounds like a sci-fi movie? Not at all! In fact, capabilities of artificial intelligence (AI) have progressed beyond what this simple example demonstrates. In China recently, a facial recognition surveillance camera network was able to apprehend a suspect from a crowd of 60,000 concert goers—possible today thanks to real-time edge analytics.

What is edge analytics

In a typical Internet of Things (IoT) system, every connected device, except for the cloud, is considered an edge device—cellphone, surveillance camera, connected automobile, Internet gateway, etc. Today, all data is digital and the number of connected devices is expected to grow to 27 billion in just three years. There has been an explosion of data generated, and the location at which data is stored and analysed has become critical to extract maximum value from it.

In the missing child scenario, surveillance video feeds from multiple cameras are sent to the cloud over 4G, analysed and then information is sent back to the authorities on the ground. Time to act is delayed and results can be tragic. Real-time analytical tools need to work with real-time data, and the best place to do that is where real-time data sits—on the edge.

Edge analytics is the lynchpin that drives real-time decision-making. It means increased intelligence on the devices at the edge. This translates to increased processing capabilities (compute) as well as higher storage on edge devices.

The ability to analyse and extract value from data in real time is a game changer for all industries. Edge analytics is being leveraged by all verticals, including smart cities, manufacturing, retail and healthcare.

Edge devices deliver timely, informed decisions

At a high level, there are three crucial functions that define an edge device—compute (processor), storage and communication. Depending on the application, there may be a need for faster processing, multi-core processors to save power, more sensors, contextual awareness and a huge storage capacity.

Till now, raw data content sat at the edge and metadata was sent to the cloud, but increasingly, context is complementing content. Both content and context drive informed real-time decisions. Raw content from the constant inflow of data from sensors must be turned readily into information or context.

Technologies like AI, machine learning (ML) and image recognition, when applied to edge devices, interact with real-time data to generate context. With context, we gain quick and actionable insights to immediate environmental stimuli. Access to this kind of information ultimately creates a more efficient and effective environment.

For instance, let us look at a home surveillance camera. Typically, when someone rings your door bell, you can determine whether it is someone you know—such as a relative or domestic help—before you open the door. By adding facial recognition software, the technology could verify whether the delivery truck driver who rang your doorbell is truly an employee of the company he claims to work for, just by tapping into an employee database. If the software could intimate you that the person is not who he claims to be, then this can serve as a timely warning that prevents you from opening the door.

Another application of edge analytics is in the automotive industry. With connected cars or fully autonomous vehicles, proximity sensors are critical to identify impending danger and play a key role in autonomous driving. These vehicles' proximity sensor data, however, may never be pushed to the cloud because these cannot afford time lag.

When a vehicle is too close to an object for comfort, analysis needs to be near-instantaneous, based on real-time data, in the car itself, to deliver a real-time response. On completing the journey however, proximity data may no longer be needed (since the environment might have changed). Thus, this kind of data could be flushed from memory, post occurrence, since it will no longer be of use.

To truly take advantage of actionable insights and real-time response on edge devices, both reliable, resilient storage and powerful analytical tools will need to reside at the edge. Here, timeliness is critical, as this actually converts content into context and passive decisions into informed ones.

Embedded solutions help capture, aggregate, transform and preserve data across IoT devices that range from smartphones, drones and surveillance cameras, to connected cars, appliances, personal devices, sensors and more. These devices reside at the edge where embedded solutions work behind the scenes in increasingly smaller, but simultaneously more powerful solutions. AI is hard at work at the edge, analysing and processing data to give real-time feedback.

What lies ahead

AI seems to be touching lives at every turn. Typically, you listen to radio traffic reports during your morning commute. You might hear about a car accident on the radio and take a different route. Today, AI-infused talking personal assistants on your smartphone can give you the best way home and even reroute you along the way.

Thanks to AI and other technologies, we know more about our surroundings than ever before. We are also starting to learn more about